How-To Library

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Using PowerExchange to Validate and Extract Data in Flat Files

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Abstract

Data in flat files must be mapped to a datatype before it can be retired. You can use PowerExchange to create data maps that specify the datatypes for data in flat files. After you create a data map, you can validate and correct the data before you retire it.

Supported Versions

- PowerExchange 10.0 or later
- Data Archive 6.4 or later

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Overview

Before you can retire data from flat files, you must first create a data map and assign a datatype to each column of data in the flat file. You must also validate the data and correct or skip rows with incorrect data.

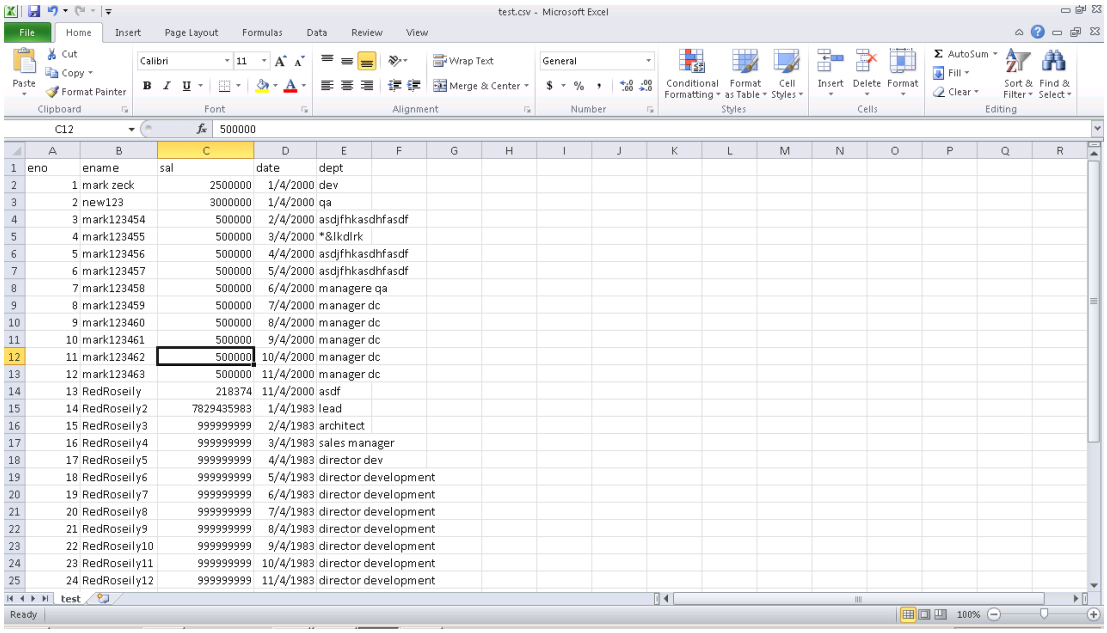
Completing these tasks manually is time consuming and prone to errors. Instead, use PowerExchange to automate and scale the data mapping process for data in one or more flat files at a time. Complete the following steps in PowerExchange to create a data map of data in flat files:

- Set up a data map
- Assign datatypes
- Add a table
- Validate datatypes
- Correct datatypes

You can map datatypes for data in more than one file at a time if the data has the same format in all the flat files in the group.

Example of a Flat File

The following figure shows sample data in a flat file:



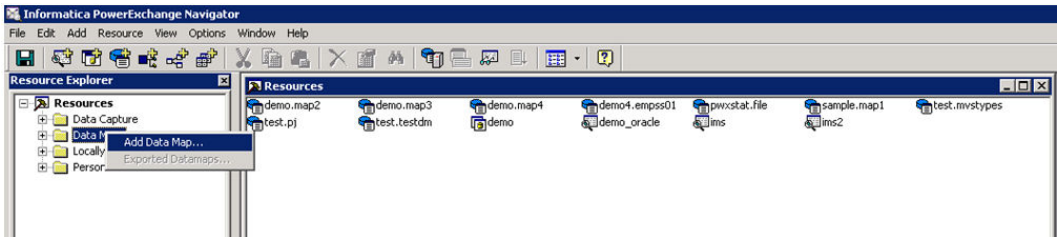
eno	ename	sal	date	dept
1	mark zeck	2500000	1/4/2000	dev
2	new123	3000000	1/4/2000	qa
3	mark123454	500000	2/4/2000	asdjfhkasdhfasdf
4	mark123455	500000	3/4/2000	*&lkdlrk
5	mark123456	500000	4/4/2000	asdjfhkasdhfasdf
6	mark123457	500000	5/4/2000	asdjfhkasdhfasdf
7	mark123458	500000	6/4/2000	managere qa
8	mark123459	500000	7/4/2000	manager dc
9	mark123460	500000	8/4/2000	manager dc
10	mark123461	500000	9/4/2000	manager dc
11	mark123462	500000	10/4/2000	manager dc
12	mark123463	500000	11/4/2000	manager dc
13	RedRoseily	218374	11/4/2000	asdf
14	RedRoseily2	7829435983	1/4/1983	lead
15	RedRoseily3	999999999	2/4/1983	architect
16	RedRoseily4	999999999	3/4/1983	sales manager
17	RedRoseily5	999999999	4/4/1983	director dev
18	RedRoseily6	999999999	5/4/1983	director development
19	RedRoseily7	999999999	6/4/1983	director development
20	RedRoseily8	999999999	7/4/1983	director development
21	RedRoseily9	999999999	8/4/1983	director development
22	RedRoseily10	999999999	9/4/1983	director development
23	RedRoseily11	999999999	10/4/1983	director development
24	RedRoseily12	999999999	11/4/1983	director development

Flat File Data Definition

To define data in a flat file, set up a data map, designate a datatype for each column in the flat file, then validate, correct, and extract data.

Step 1. Set Up a Data Map

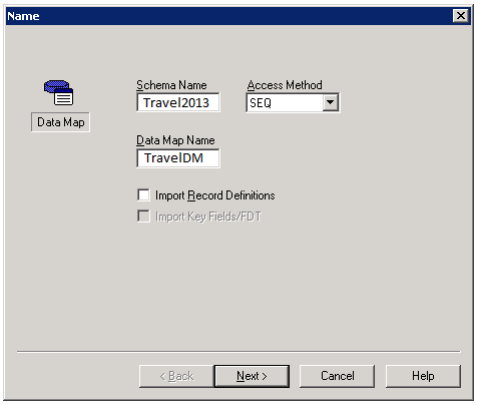
1. Open the PowerExchange Navigator.
2. On the **Resource Explorer** pane, right-click the **Data Map** folder and select **Add Data Map**.



   The **Name** dialog box appears.

3. Enter the following attributes for the data map:
   a. Enter a name for the schema.
   b. Select **SEQ** from the **Access Method** list.
   c. Enter a name for the data map.

d. Clear the **Import Record Definitions** check box.



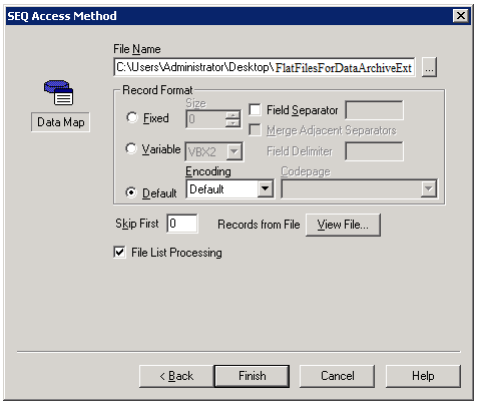
4. Click **Next**.

   The **SEQ Access Method** dialog box appears.

5. Use the following steps to configure the file attributes:

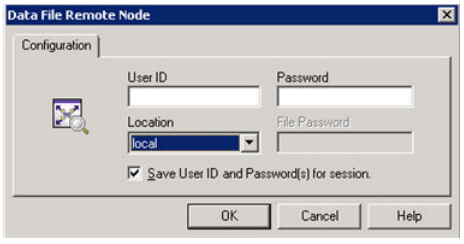
   a. Use the browse button in **File Name** to select the file path.

   b. Optionally, select the field separator.



   c. Click **View File** to view the records in the flat file.

   The **Data File Remote Node** dialog box appears.



6. Enter the following information to access the flat file:

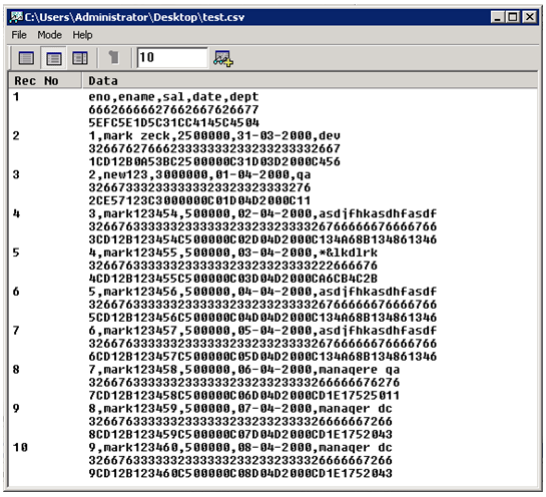
   a. Enter the **User ID**.

   b. Optionally, enter the **Password**.

   c. Enter **local** as the **Location** if you have the flat file saved locally on your computer.

d. Click **OK**.

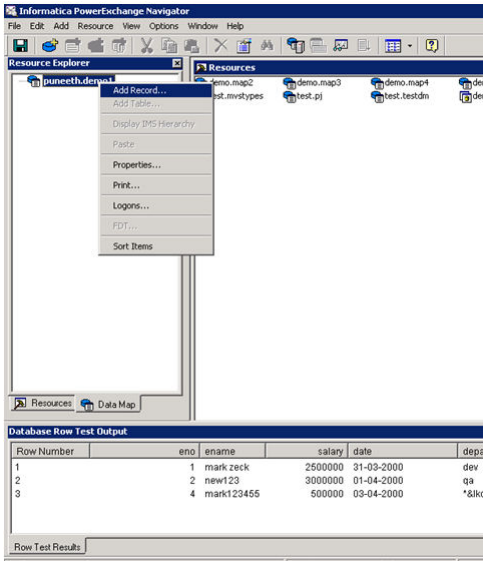
The flat file appears.



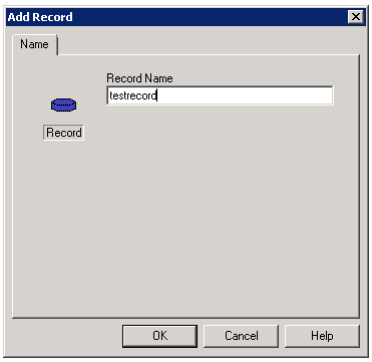
7. Review and close the file.
8. Click **Finish** on the **SEQ Access Method** dialog box.

Step 2. Assign Datatypes

1. On the **Resource Explorer** pane, right-click the data map and select **Add Record** .



The **Add Record** dialog box appears.



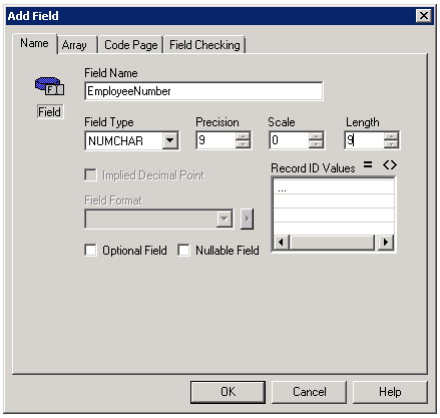
2. Enter a name for the record and click **OK**

   The **Record - <record name>** window appears.

3. Right-click the record and select **Add Field at End**.



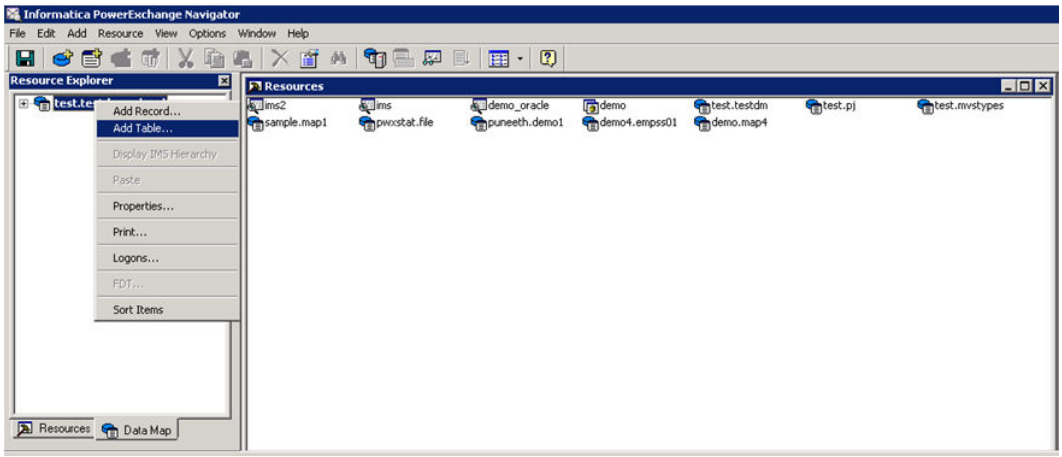
The **Add Field** dialog box appears.



4. Enter a column or field name.
5. Select the datatype, precision, scale, and length.
6. Click **OK**.
7. Repeat steps 3 through 6 for each column in the flat file.

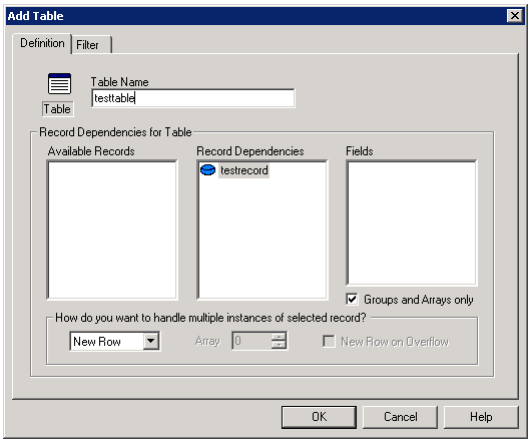
Step 3. Add Table and Validate Data

1. Right-click the data map and select **Add Table** .

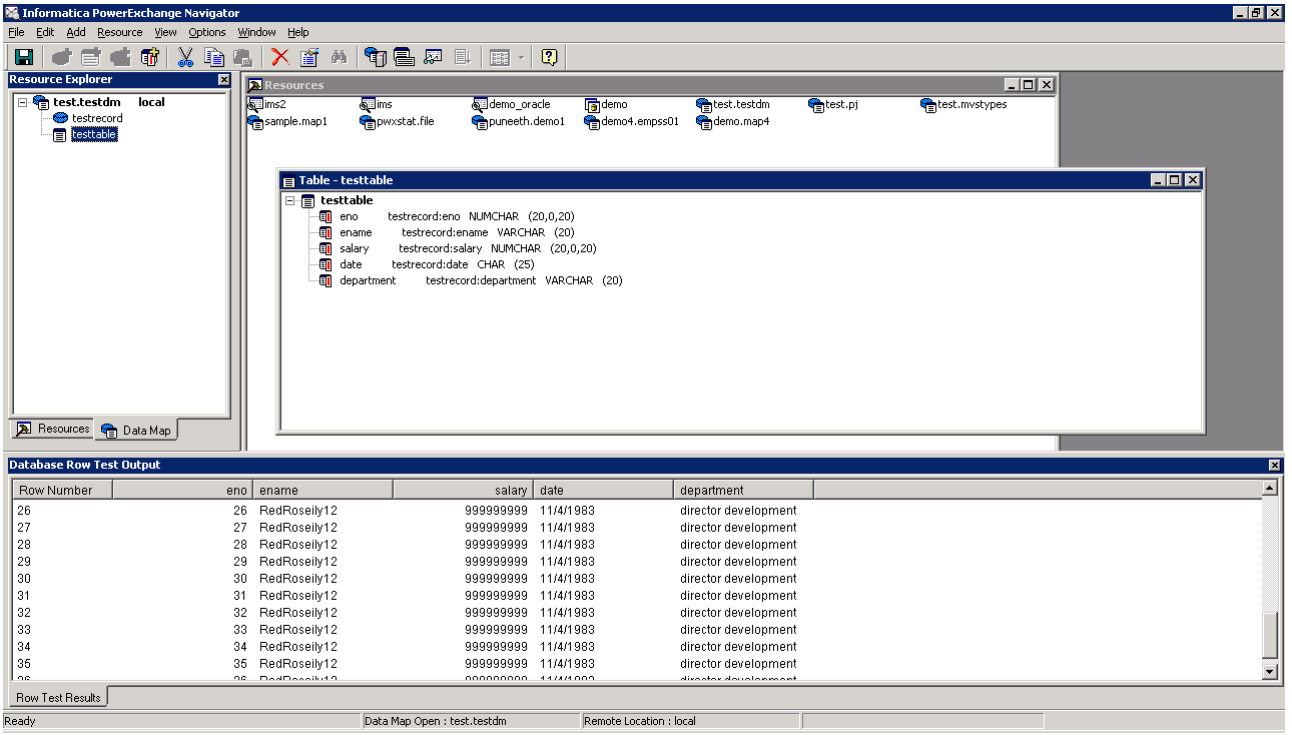


The **Add Table** dialog box appears.

2. Enter the table name and click **OK**.



The **Table - <table name>** window appears.



3. Right-click the table name and select **Properties** .

   The **Data Map Properties** dialog box appears.

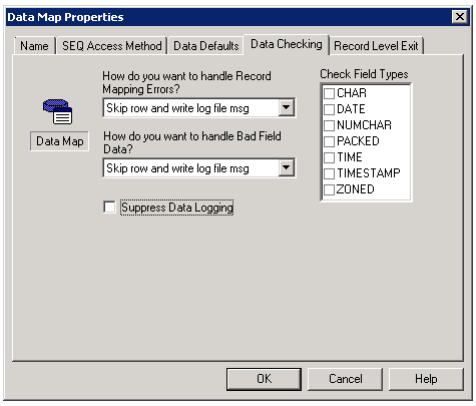
4. Open the **Data Checking** tab.
5. Select **Terminate the data extract** in the event of record-mapping errors.
6. Select **Terminate the data extract** in the event of incorrect field data.

Step 4. Correct and Extract Data

You can choose one of two ways to correct errors in the data.

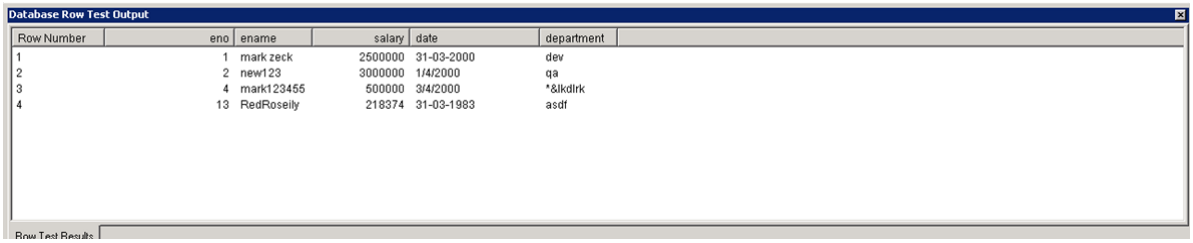
Option 1. Skip Rows with Errors

1. Go to the Data Map Properties dialog box and open the Data Checking tab.
2. Select **Skip row and write log file msg** in the event of record-mapping errors
3. Select **Skip row and write log file msg** in the event of incorrect field data.
4. Enable field types.



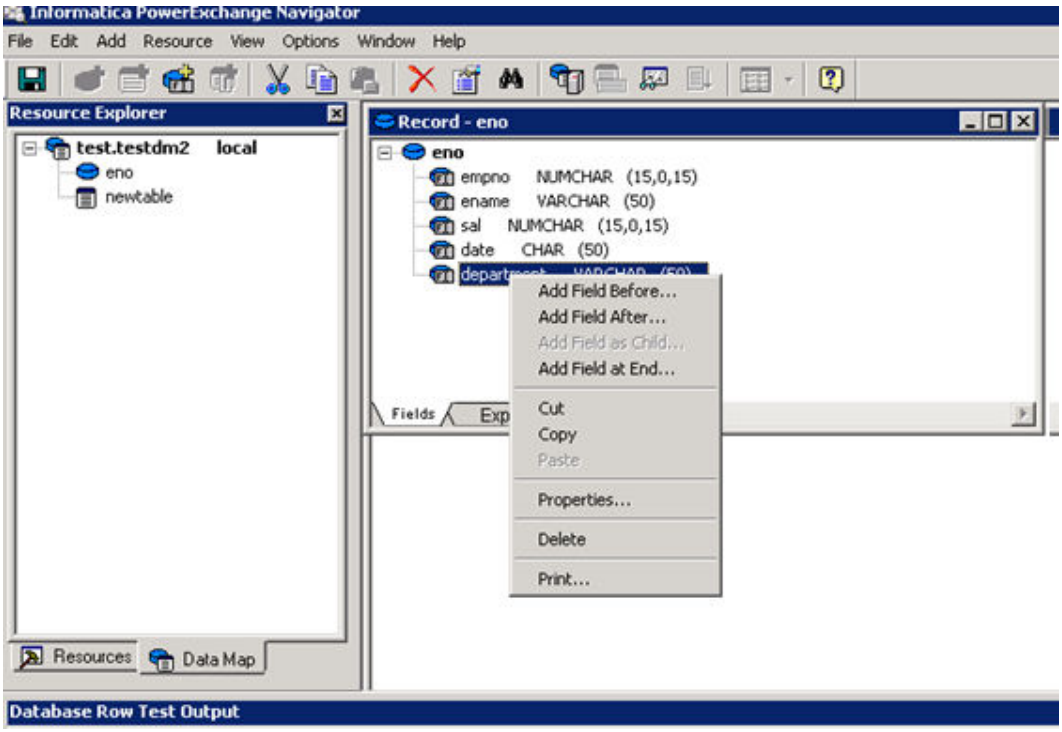
5. Click **OK** to validate.

   A list of records containing the correct data that match the datatype appear. Records with bad data or data mapping errors are omitted.



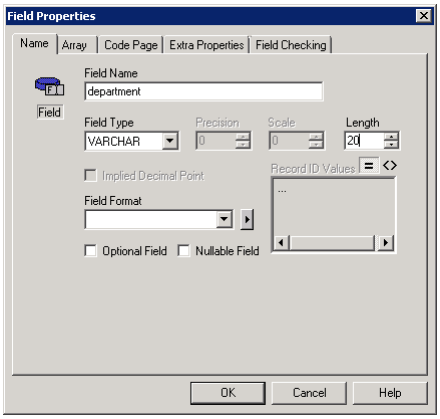
Option 2. Change Record Properties

1. From the **Database Row Test Output** window, right-click the record with the error and select **Properties**.



   The **Field Properties** dialog box appears.

2. Edit the properties as required. For example, if the field length exceeds the datatype length, increase the length of that datatype



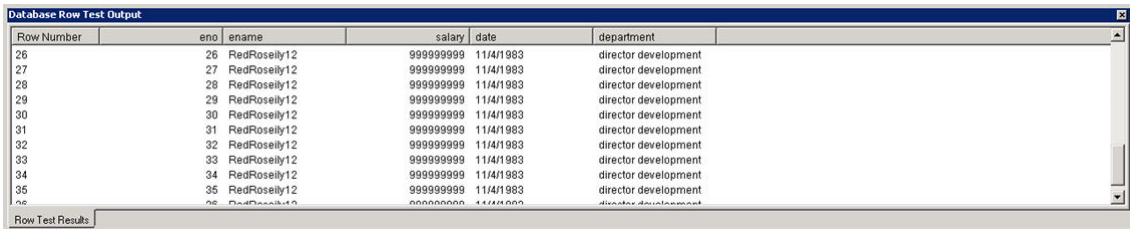
3. Click **OK**.
4. Go to the **Resource Explorer** view.
5. Expand the Data Map folder.
6. Select the table and click the **Database Row Test** icon to validate the data map.

   The **Database Row Test** dialog box appears.

7. Select the default database type.
8. Select **local** for **Location**.

9. Select the number of rows in the **Get Rows** field.
10. Click Go.

    The **Database Row Test Output** page appears with the records of the flat file.

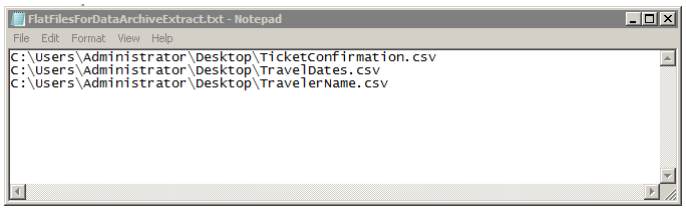


Define Data in Multiple Flat Files at a Time

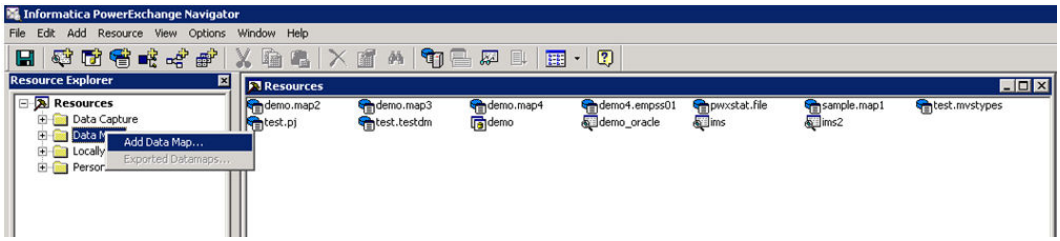
You can create a data map for multiple flat files at a time if all the flat files in the group have the same data format.

1. List the file path and file names of the flat files in a text file.

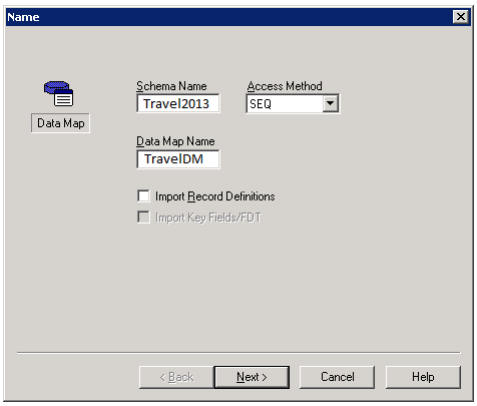
   The following image shows an example text file with a list of flat files:



2. Open PowerExchange Navigator.
3. On the **Resource Explorer** pane, right-click the Data Map folder and select **Add Data Map**.

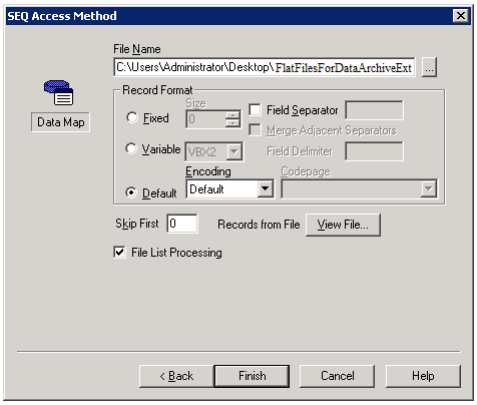


   The **Name** dialog box appears.



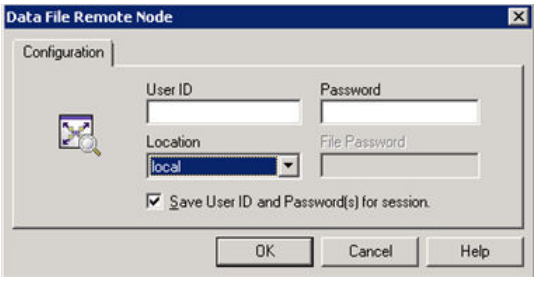
4. Complete the following information in the **Name** dialog box:
   a. Enter a **Schema Name**.
   b. Select **SEQ** from the **Access Method** list.
   c. Enter **Data Map Name**.
   d. Disable **Import Record Definitions**.
5. Click **Next**.

   The **SEQ Access Method** dialog box appears.



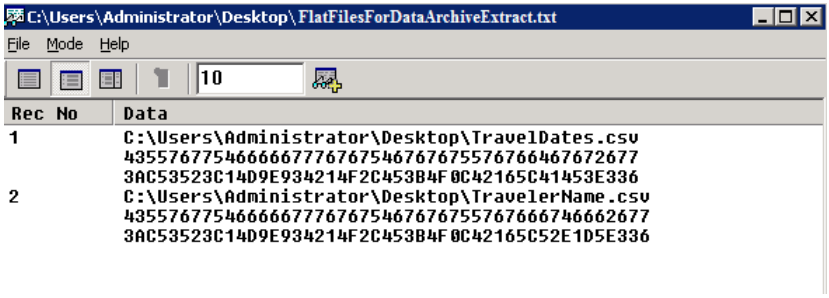
6. Complete the following information in the **SEQ Access Method** dialog box:
   a. Use the browse button in **File Name** to select the text file that contains the list of flat files.
   b. Optionally, select the **Field Separator** check box.
   c. Enable **File List Processing**.
7. Click **View File** to open the text file that contains the list of flat files

   The **Data File Remote Node** dialog box appears.



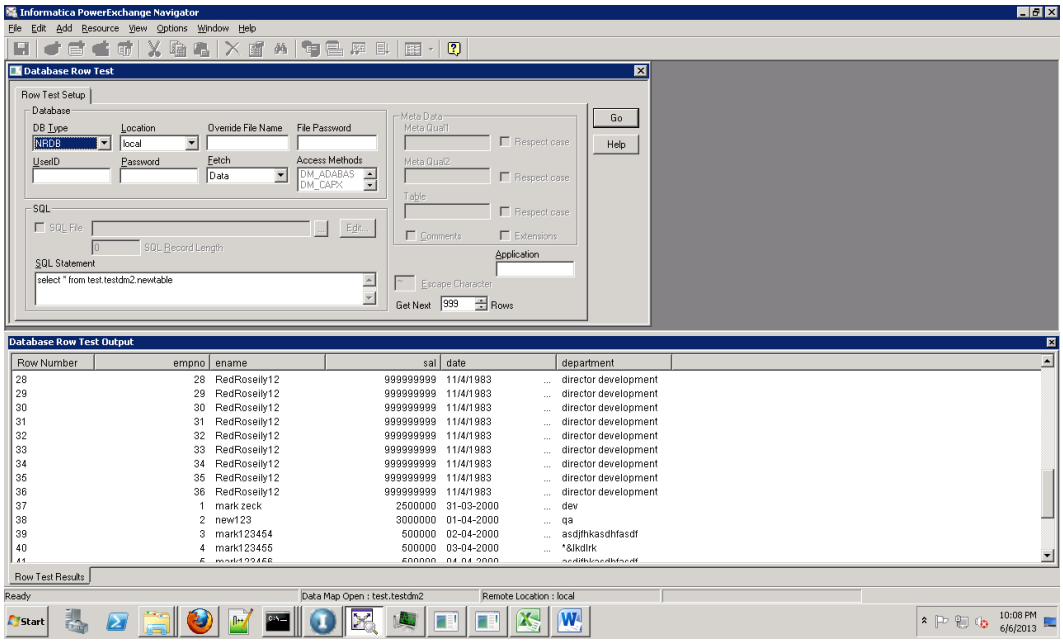
8. Complete the following information in the **Data File Remote Node** dialog box:
   a. Enter the **User ID**.
   b. Optionally, enter the **Password**.
   c. Enter **local** as the location if you have the flat file saved locally on your computer.
9. Click **OK**

The text file with the list of flat files appears.



10. Review and close the file.
11. Click **Finish** on the **SEQ Access Method** dialog box.
12. Map column datatypes. For more information, see "Step 2. Assign Datatypes" on page 6.
13. Add a table and validate data. For more information, see "Step 3. Add Table and Validate Data" on page 7.
14. Correct and extract data. For more information, see "Step 4. Correct and Extract Data" on page 9.

    The **Database Row Test Output** window displays the records from the flat files in the group.



Conclusion

After you create the data map, retire the data to File Archive Service in the same way you retire legacy applications.

Authors

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